# SELMER ELLIPTIC CURVES: SOLUTIONS

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The problems below study a single family of plane cubics and use it to reconstruct, by hand, some of the main results of [Sil09, §§III.1–III.4]: Weierstrass equations, the group law, the Riemann–Roch construction of a Weierstrass model, and a first endomorphism. Section 4 is a set of worked `Magma` examples to read and re-run afterwards, with three short variations to try at the end. Each problem is preceded by the definitions and results it uses.

**Standing notation.**

- A field $k$ with $\operatorname{char} k \neq 3$, an algebraic closure $\overline{k}$ of $k$, and $d \in k^\times$.
- The cubic form $F := X^3 + Y^3 - dZ^3$, the plane cubic $C_d := \{F = 0\} \subset \mathbb{P}^2_k$, and its rational point $O := (1 : -1 : 0)$.
- A primitive cube root of unity $\zeta \in \overline{k}$.
- A Selmer elliptic curve: a pair $(C_d, O)$ of this shape, after [Sel51]. Section 5 explains the name.
- Projective coordinates $X, Y, Z$, as usual, and lowercase letters for affine coordinates.
- The base change $\overline{C}_d := C_d \times_k \overline{k}$.

**How to use this.** Sections 1–2 are meant for the problem session. Section 3 may be started there and finished at home. As a rough guide, allow about 12 minutes for §1 and 33 for §2. Within §2, Problem 2.2 is the core and Problem 2.5(5)–(7) the payoff. If time is short, leave Problem 2.5(1)–(4) and Problem 2.8 for later. Section 4 needs no time in the session. It is worked examples, not exercises.

## 1. Warm-up: a plane cubic and a rational point on it

**Definition 1.1** (Elliptic curves [Sil09, §III.3])**.** An elliptic curve over $k$ is a pair $(E, O)$ with $E$ a smooth, projective, geometrically irreducible curve of genus one over $k$ and $O \in E(k)$ a fixed rational point.

**Problem 1.2** (It is an elliptic curve)**.**

(1) Check that $O \in C_d(k)$.

Solution. $1^3 + (-1)^3 - d \cdot 0 = 0$.

(2) Compute the three partial derivatives of $F$ and deduce that $C_d$ is smooth, using $\operatorname{char} k \neq 3$ and $d \neq 0$. Where is each hypothesis used?

Solution. **$\operatorname{char} k \neq 3$ to cancel the 3, and $d \neq 0$ to cancel $d$.**

$$\partial F/\partial X = 3X^2, \qquad \partial F/\partial Y = 3Y^2, \qquad \partial F/\partial Z = -3dZ^2.$$

If all three vanish at a point of $\mathbb{P}^2(\overline{k})$ then $3X^2 = 3Y^2 = 3dZ^2 = 0$. Since $3 \neq 0$ in $k$ we get $X^2 = Y^2 = 0$, and since also $d \neq 0$ we get $Z^2 = 0$. So $X = Y = Z = 0$, which is not a point of $\mathbb{P}^2(\overline{k})$. Both hypotheses are used to divide: $\operatorname{char} k \neq 3$ to cancel the 3, and $d \neq 0$ to cancel $d$ in the third equation.

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*Remark.* If $d = 0$ the "curve" $X^3 + Y^3 = 0$ is three concurrent lines.

(3) Suppose instead that $\operatorname{char} k = 3$ (only for this problem). Show that over $\overline{k}$ the curve $C_d$ is a *triple line*.

Solution. In characteristic three the Frobenius $a \mapsto a^3$ is a ring homomorphism, so for $\delta \in \overline{k}$ with $\delta^3 = d$,

$$(X + Y - \delta Z)^3 = X^3 + Y^3 + (-\delta Z)^3 = X^3 + Y^3 - dZ^3 = F.$$

Thus $\overline{C}_d$ is the triple line $\{X + Y - \delta Z = 0\}$: not reduced, hence certainly not smooth, and of arithmetic genus 1 for entirely uninteresting reasons.

*Remark.* Note that for $d \notin (k^\times)^3$ the polynomial $F$ is still irreducible *over* $k$, so the failure is invisible until one passes to $\overline{k}$.

(4) Show that a smooth plane curve is geometrically irreducible.

Solution. If a plane curve $C$ decomposes as $C_1 \cup C_2$ with $C_i$ of degrees $e_i \geqslant 1$, then by Bézout $(C_1 \cap C_2)(\overline{k}) \neq \varnothing$, and any point of $(C_1 \cap C_2)(\overline{k})$ is a singular point of $C$. So a smooth plane curve is irreducible. Applied over $\overline{k}$ this gives geometric irreducibility.

(5) Conclude from the degree–genus formula that $(C_d, O)$ is an elliptic curve over $k$.

Solution. A smooth plane curve of degree $m$ has genus $(m-1)(m-2)/2$. For $m = 3$ this is 1. With (1), (2) and (4), the pair $(C_d, O)$ is an elliptic curve over $k$.

The group law in Problem 1.4(4) is the chord–tangent construction of [Sil09, §III.2].

**Proposition 1.3** (The group law and $\operatorname{Pic}^0$ [Sil09, Proposition III.3.4])**.** *The map $P \mapsto [P - O]$ identifies the chord–tangent group law on $C_d(\overline{k})$ with addition in $\operatorname{Pic}^0 \overline{C}_d$.*

**Problem 1.4** (The base point is an inflection point)**.**

(1) Show that the line $Z = 0$ meets $C_d$ in three *distinct* points of $C_d(\overline{k})$, and write them down. For which $k$ are all three in $C_d(k)$?

Solution. **All three lie in $C_d(k)$ exactly when $\zeta \in k$.** Setting $Z = 0$ gives $X^3 = -Y^3$, so $X/Y \in \{-1, -\zeta, -\zeta^2\}$ and the points of $C_d(\overline{k})$ on the line $Z = 0$ are

$$O = (1 : -1 : 0), \qquad Q_1 = (1 : -\zeta : 0), \qquad Q_2 = (1 : -\zeta^2 : 0).$$

They are distinct because $T^3 + 1$ is separable when $\operatorname{char} k \neq 3$.

(2) Compute the tangent line $L$ to $C_d$ at $O$, and show that $L \cap C_d = 3O$. (So $O$ is an inflection point, or flex.)

Solution. $\boldsymbol{L = \{X + Y = 0\}}$**.** From Problem 1.2(2), $\nabla F(O) = (3, 3, 0)$, so $L = \{X + Y = 0\}$. Substituting $X = -Y$ into $F$ gives $(-Y)^3 + Y^3 - dZ^3 = -dZ^3$, which vanishes to order three at $Z = 0$. Hence $L \cap C_d = 3O$.

(3) In a Weierstrass model, which line plays the role of $L$? What is different here?

Solution. **The line at infinity $Z = 0$.** In a Weierstrass model it is the line at infinity $Z = 0$ that cuts $3O$. Here $Z = 0$ does something different: by (1) it meets $C_d$ in three distinct points of $C_d(\overline{k})$. The flex tangent is instead $X + Y = 0$.

*Remark.* It is worth noticing that $O$ being a flex is what makes the group law take the symmetric form in (4). It is not automatic for a plane cubic with a marked rational point.

(4) Let $\oplus$ denote the group law on $C_d$ with identity $O$, defined by the chord–tangent construction. Using (2), show that for $P, Q, R \in C_d(\overline{k})$,

$$P \oplus Q \oplus R = O \iff P + Q + R \text{ is cut out on } C_d \text{ by a line.}$$

Solution. Write $\ell$ for a linear form cutting $P+Q+R$ and $\ell_O = X + Y$ for the flex tangent. Then $\operatorname{div}(\ell/\ell_O) = (P+Q+R) - 3O$ in $\operatorname{Div} \overline{C}_d$, so $P+Q+R \sim 3O$. Under the isomorphism $C_d(\overline{k}) \xrightarrow{\sim} \operatorname{Pic}^0 \overline{C}_d$, $P \mapsto [P - O]$, we get

$$P \oplus Q \oplus R \longmapsto [P + Q + R - 3O] = 0,$$

so $P \oplus Q \oplus R = O$. Conversely, suppose $P \oplus Q \oplus R = O$. The line through $P$ and $Q$ meets $C_d$ in a third point $R' \in C_d(\overline{k})$, and the direction just proved gives $P \oplus Q \oplus R' = O$. So $R' = R$, and $P + Q + R$ is cut out by that line.

## 2. The group law

**Problem 2.1** (Negation, and 2-torsion)**.**

(1) Show that $\ominus(X_0 : Y_0 : Z_0) = (Y_0 : X_0 : Z_0)$ for every point $(X_0 : Y_0 : Z_0) \in C_d(\overline{k})$. There are two things to check: that the right-hand side lies on $C_d$, and that it is the third intersection point of $C_d$ with the line through $O$ and $(X_0 : Y_0 : Z_0)$.

Solution. $F$ is symmetric in $X$ and $Y$, so $P' \coloneqq (Y_0 : X_0 : Z_0) \in C_d(\overline{k})$. Write $P \coloneqq (X_0 : Y_0 : Z_0)$. For collinearity,

$$\det \begin{pmatrix} 1 & -1 & 0 \\ X_0 & Y_0 & Z_0 \\ Y_0 & X_0 & Z_0 \end{pmatrix} = 1 \cdot (Y_0 Z_0 - X_0 Z_0) - (-1) \cdot (X_0 Z_0 - Y_0 Z_0) + 0 = 0.$$

If $O$, $P$, $P'$ are distinct, the line through them therefore cuts out exactly $O + P + P'$. Otherwise $P' = P$ (note that $P' = O$ only when $P = O$): if $Z_0 \neq 0$ this means $X_0 = Y_0$, and if $Z_0 = 0$ it means $X_0 = \pm Y_0$, and of the three $\overline{k}$-points of Problem 1.4(1) only $O$ qualifies. For $P = O$ the flex tangent cuts out $3O$ by Problem 1.4(2). For $X_0 = Y_0$ the two last rows coincide and the determinant says nothing, but the tangent line to $C_d$ at $P$, $X_0^2 X + X_0^2 Y - d Z_0^2 Z = 0$, passes through $O$. So the line through $O$ and $P$ is tangent at $P$, and it cuts out $O + 2P$. In every case $O + P + P'$ is cut out by a line, and since $O$ is a flex, Problem 1.4(4) gives

$$O \oplus P \oplus P' = O, \quad \text{i.e.} \quad (Y_0 : X_0 : Z_0) = \ominus(X_0 : Y_0 : Z_0).$$

(2) Deduce that the nontrivial 2-torsion of $C_d$ is cut out by the line $X = Y$, and that $\#C_d[2](\overline{k}) = 4$ when $\operatorname{char} k \neq 2$.

Solution. For $P = (X_0 : Y_0 : Z_0) \in C_d(\overline{k})$, $P = \ominus P$ means $(X_0 : Y_0 : Z_0) = (Y_0 : X_0 : Z_0)$ in $\mathbb{P}^2(\overline{k})$. If $Z_0 \neq 0$, normalize $Z_0 = 1$: then $X_0 = Y_0$ and $2X_0^3 = d$, which has three solutions in $\overline{k}$ since $\operatorname{char} k \neq 2$. If $Z_0 = 0$ then $P$ is one of the three $\overline{k}$-points of Problem 1.4(1), and $(X_0 : Y_0 : 0) = (Y_0 : X_0 : 0)$ means $X_0^2 = Y_0^2$, i.e. $Y_0/X_0 = \pm 1$. Of $-1, -\zeta, -\zeta^2$ only $-1$ qualifies, so $P = O$. Hence the nontrivial 2-torsion consists of the three points of $C_d(\overline{k})$ on the line $X = Y$, and $\#C_d[2](\overline{k}) = 3 + 1 = 4$, as it must be for an elliptic curve in characteristic $\neq 2$.

(3) Take $k = \mathbb{Q}$. For which $d$ does $C_d$ have a nontrivial rational 2-torsion point? Write it down, and give the smallest positive integer $d$ for which this happens.

SOLUTION. **$d = 2c^3$ with $c \in \mathbb{Q}^\times$. Smallest: $d = 2$.** A nontrivial rational 2-torsion point exists if and only if $d/2 \in (\mathbb{Q}^\times)^3$, say $d = 2c^3$. It is then $(c : c : 1)$. The smallest positive integer is $d = 2$, with the rational point $(1 : 1 : 1)$, i.e. $1^3 + 1^3 = 2$.

Part (2) of Problem 2.2 is its heart. Parts (4) and (5) are then mechanical. If time is short, take (2) on faith and come back to its proof, but do not skip (3): it is what makes (5) painless. The identity of (2) is the polarization of a cubic form. For a diagonal cubic the trilinear form can be written down by hand.

**Problem 2.2** (The trilinear form, and the addition formulas)**.**
For vectors $P = (X_P, Y_P, Z_P)$, $Q$ and $R$ in $\overline{k}^3$, define

$$\langle P, Q, R\rangle := X_P X_Q X_R + Y_P Y_Q Y_R - d\, Z_P Z_Q Z_R. \tag{2.1}$$

(1) Check that (2.1) is symmetric in $P, Q, R$, linear in each argument separately, and satisfies $\langle P, P, P\rangle = F(P)$.

SOLUTION. All three properties are visible: each monomial contains exactly one coordinate from each of $P, Q, R$, so the expression is linear in each argument; permuting $P, Q, R$ permutes the factors within each monomial; and setting $P = Q = R$ gives $X_P^3 + Y_P^3 - dZ_P^3 = F(P)$.

(2) Prove the *expansion identity*: for scalars $s, t$,

$$F(sP + tQ) = s^3\langle P, P, P\rangle + 3s^2t\,\langle P, P, Q\rangle + 3st^2\,\langle P, Q, Q\rangle + t^3\langle Q, Q, Q\rangle. \tag{2.2}$$

Where do the 3's come from?

SOLUTION. **The 3's are binomial coefficients.** Expanding $\langle sP + tQ, sP + tQ, sP + tQ\rangle$ by linearity in each slot gives $2^3 = 8$ terms, indexed by which of $sP$, $tQ$ goes into each slot. By the symmetry of (1), the value depends only on *how many* slots receive $tQ$: there are $\binom{3}{1} = 3$ terms with one $tQ$, each equal to $s^2t\,\langle P, P, Q\rangle$, and $\binom{3}{2} = 3$ terms with two.

(3) Prove that $3\langle P, P, Q\rangle = \nabla F(P) \cdot Q$, and deduce that for $P \in C_d(\overline{k})$,

$$\langle P, P, Q\rangle = 0 \iff Q \text{ lies on the tangent line to } C_d \text{ at } P.$$

SOLUTION. Both sides equal

$$3(X_P^2 X_Q + Y_P^2 Y_Q - dZ_P^2 Z_Q),$$

as $\nabla F(P) = (3X_P^2,\, 3Y_P^2,\, -3dZ_P^2)$. The tangent line to $C_d$ at a smooth point $P$ is $\{\nabla F(P) \cdot (X, Y, Z) = 0\}$, so $\langle P, P, Q\rangle = 0$ says exactly that $Q$ lies on it.

(4) Let $P_1 = (X_1 : Y_1 : Z_1)$ and $P_2 = (X_2 : Y_2 : Z_2)$ be distinct points of $C_d(\overline{k})$, and parametrize the line through them as $sP_1 + tP_2$. Using (2), show that the third intersection point is at $(s : t) = (B : -A)$ where $A = \langle P_1, P_1, P_2\rangle$ and $B = \langle P_1, P_2, P_2\rangle$, and hence that

$$P_1 \oplus P_2 = (BY_1 - AY_2 : BX_1 - AX_2 : BZ_1 - AZ_2). \tag{2.3}$$

Why can $A$ and $B$ not both vanish?

SOLUTION. Since $F(P_1) = F(P_2) = 0$, (2.2) reduces to

$$F(sP_1 + tP_2) = 3s^2t\,A + 3st^2\,B = 3st(As + Bt).$$

This binary cubic in $(s : t)$ has roots $(1 : 0)$, $(0 : 1)$ and $(B : -A)$. The first two are $P_1$ and $P_2$, so the third intersection point is $BP_1 - AP_2$. Composing with $\ominus$ from Problem 2.1(1), which

swaps the first two coordinates, gives (2.3). If $A = B = 0$ then $F$ vanishes identically on the line, so the line would be a component of $C_d$, contradicting irreducibility (Problem 1.2(4)).

(5) Let $P = (X_0 : Y_0 : Z_0) \in C_d(\overline{k})$ and put $Q := (Y_0^2 : -X_0^2 : 0)$. Verify that $Q$ lies on the tangent to $C_d$ at $P$, and use (2) and (3) to derive

$$[2]P = \big(-Y_0(2X_0^3 + Y_0^3) : X_0(X_0^3 + 2Y_0^3) : Z_0(X_0^3 - Y_0^3)\big). \tag{2.4}$$

Solution. $\langle P, P, Q\rangle = X_0^2Y_0^2 + Y_0^2(-X_0^2) - dZ_0^2 \cdot 0 = 0$, so $Q$ is on the tangent by (3). Now (2.2) loses two terms:

$$F(sP + tQ) = 3st^2\,\langle P, Q, Q\rangle + t^3F(Q).$$

The double root at $t = 0$ is the tangency at $P$. The third root gives the third intersection point of the tangent with $C_d$, call it $P * P$:

$$P * P = -F(Q)\,P + 3\langle P, Q, Q\rangle\,Q.$$

Here

$$F(Q) = Y_0^6 - X_0^6, \qquad \langle P, Q, Q\rangle = X_0Y_0^4 + Y_0X_0^4 = X_0Y_0(X_0^3 + Y_0^3) = dX_0Y_0Z_0^3.$$

Substituting and using $dZ_0^3 = X_0^3 + Y_0^3$ to simplify, each coordinate acquires the factor $X_0^3 + Y_0^3$. Cancelling it and applying $\ominus$ gives (2.4). Explicitly, the first coordinate before cancelling is

$$X_0(X_0^6 - Y_0^6) + 3dX_0Y_0^3Z_0^3 = X_0(X_0^3 + Y_0^3)(X_0^3 + 2Y_0^3).$$

The cancellation needs $X_0^3 + Y_0^3 = dZ_0^3 \neq 0$, that is, $Z_0 \neq 0$. On the line $Z = 0$ the derivation says nothing: there $Y_0^3 = -X_0^3$, so $Q = (Y_0^2 : -X_0^2 : 0) = (Y_0^3 : -X_0^2Y_0 : 0) = (X_0 : Y_0 : 0)$ is $P$ itself and $F(sP + tQ)$ vanishes identically. But (2.4) holds there too. The tangent line at $P = (X_0 : Y_0 : 0)$ is $X_0^2X + Y_0^2Y = 0$. On it $X^3 = -Y_0^6Y^3/X_0^6 = -Y^3$, so $F$ restricts to $-dZ^3$ and the tangent cuts out $3P$, exactly as at $O$ in Problem 1.4(2). Hence $P * P = P$ and $[2]P = \ominus P = (Y_0 : X_0 : 0)$, while the right-hand side of (2.4) is $(-Y_0X_0^3 : -X_0^4 : 0) = (Y_0 : X_0 : 0)$.

(6) Check (2.4) against Problem 2.1: what does it give on the line $X = Y$? (The line $X = 0$ is taken up in Problem 2.5.)

Solution. **$[2]P = O$.** If $X_0 = Y_0$, then $[2]P = (-3X_0^4 : 3X_0^4 : 0) = (1 : -1 : 0) = O$, consistent with Problem 2.1(2).

(7) Where does the whole method break down if $\operatorname{char} k = 3$? Compare your answer with Problem 1.2(3).

Solution. **The middle coefficients of (2.2) are literally 3.** In characteristic three the identity collapses to $F(sP + tQ) = s^3F(P) + t^3F(Q)$, which vanishes identically once $P$ and $Q$ represent points of $C_d(\overline{k})$: every line "meets" the cubic everywhere and the chord–tangent construction carries no information at all. This is the same degeneration as in Problem 1.2(3), seen from the side of the group law.

**Problem 2.3** (Arithmetic on $C_7$)**.**
Let $d = 7$, so that $P := (2 : -1 : 1) \in C_7(\mathbb{Q})$ because $2^3 + (-1)^3 = 7$.

(1) Compute $[2]P$ using (2.4), and simplify. Verify your answer by exhibiting an identity between cubes of integers.

SOLUTION. **$[2]P = (5 : 4 : 3)$.** With $X_0 = 2$, $Y_0 = -1$, $Z_0 = 1$ we have $X_0^3 = 8$ and $Y_0^3 = -1$, so

$$[2]P = \big(-(-1)(16-1) : 2(8-2) : 1\cdot(8+1)\big) = (15 : 12 : 9) = (5 : 4 : 3),$$

and indeed $5^3 + 4^3 = 125 + 64 = 189 = 7 \cdot 27 = 7 \cdot 3^3$.

(2) Compute $[3]P$ using (2.3) applied to $P$ and $[2]P$, and verify it the same way.

SOLUTION. **$[3]P = (-17 : 73 : 38)$.** With $P_1 = (2, -1, 1)$ and $P_2 = (5, 4, 3)$,

$$A = 4\cdot 5 + 1\cdot 4 - 7\cdot 1\cdot 3 = 20 + 4 - 21 = 3, \qquad B = 2\cdot 25 + (-1)\cdot 16 - 7\cdot 1\cdot 9 = 50 - 16 - 63 = -29.$$

Then (2.3) gives

$$[3]P = (-29(-1) - 3\cdot 4 \ : \ -29\cdot 2 - 3\cdot 5 \ : \ -29 - 9) = (17 : -73 : -38) = (-17 : 73 : 38),$$

and $73^3 - 17^3 = 389017 - 4913 = 384104 = 7 \cdot 54872 = 7 \cdot 38^3$.

(3) Compute $[4]P$.

SOLUTION. **$[4]P = (-1256 : 1265 : 183)$.** Doubling $(5 : 4 : 3)$: $X_0^3 = 125$, $Y_0^3 = 64$, so

$$[4]P = \big(-4(250 + 64) : 5(125 + 128) : 3(125 - 64)\big) = (-1256 : 1265 : 183),$$

and $1265^3 - 1256^3 = 7 \cdot 183^3$.

(4) The three rational points you have found have numerators and denominators of rapidly increasing size. What does this suggest about the order of $P$ in $C_7(\mathbb{Q})$? (You are not asked to prove it.)

SOLUTION. **That $P$ has infinite order.** If $P$ were a torsion point, its multiples would run through a finite set, so their heights would be bounded. They visibly are not. So $P$ has infinite order, and $C_7(\mathbb{Q})$ has positive rank.

*Remark.* In fact the rank is one and $P$ generates modulo torsion. For the heights behind this argument, see [Sil09, §VIII.9] or [HS00, Part B].

In Problem 2.5, parts (1)–(3) are the general theory and (5)–(7) the payoff for our family. If time is short, take (3) on faith and go to (5)–(7). Part (4) is the most satisfying: two completely different counts turn out to agree. Part (1) uses Bézout's theorem [Har77, §I.7].

**Proposition 2.4** (The Hessian criterion)**.** *Let $C = V(F) \subset \mathbb{P}^2_{\overline{k}}$ be a smooth plane curve of degree $m$ over a field of characteristic 0 or $> 3$, and let*

$$H(F) = \det\left(\frac{\partial^2 F}{\partial X_i\, \partial X_j}\right)_{0\leqslant i,j\leqslant 2}.$$

*Then a point $P \in C(\overline{k})$ is an inflection point if and only if $H(F)(P) = 0$. This is classical. Characteristic three is taken up in Problem 2.5(8) below.*

**Problem 2.5** (Inflection points and 3-torsion)**.**

(1) Let $C = V(F)$ be as in Proposition 2.4. Show that $H(F)$ is a form of degree $3(m-2)$, so of degree 3 when $m = 3$. Assuming $C$ is a smooth plane cubic, explain why $C \not\subseteq V(H(F))$, and conclude from Bézout's theorem that $C$ has exactly 9 inflection points *counted with multiplicity*.

SOLUTION. Each entry $\partial^2 F/\partial X_i \partial X_j$ is a form of degree $m-2$, and the determinant of a $3\times 3$ matrix is a sum of products of three entries, so $H(F)$ has degree $3(m-2)$. For $m = 3$ this is 3. If $C \subseteq V(H(F))$ then by Proposition 2.4 every point of $C(\overline{k})$ would be a flex, which is false

for a smooth cubic: a curve all of whose $\overline{k}$-points are flexes is a line (and (2)–(4) below show a smooth cubic has only nine). Hence $C$ and $V(H(F))$ share no component and Bézout applies, giving $\deg(C \cap V(H(F))) = 3 \cdot 3 = 9$.

(2) Now forget the Hessian. Let $C$ be a smooth plane cubic over $\overline{k}$ and suppose $O \in C(\overline{k})$ is a flex, so that $(C, O)$ is an elliptic curve with the group law of Problem 1.4(4). Show that

$$P \text{ is a flex} \iff [3]P = O,$$

so that the set of flexes is exactly $C[3](\overline{k})$.

Solution. By definition $P$ is a flex when the tangent line at $P$ meets $C$ at $P$ with multiplicity three, i.e. when the tangent cuts out the divisor $3P$. By Problem 1.4(4) that says $P \oplus P \oplus P = O$, i.e. $[3]P = O$.

(3) Show that a smooth plane cubic over $\overline{k}$ always *has* a flex, without using the Hessian. Deduce that, with no flex singled out, the set of flexes is a torsor under $\mathrm{Pic}^0(C)[3]$. In particular the number of flexes is $\#C[3](\overline{k})$.

Solution. Fix any $P_0 \in C(\overline{k})$ and consider the degree-zero class $\mathcal{O}_C(1) - 3[P_0] \in \mathrm{Pic}^0 C$ (here $C$ is already over $\overline{k}$, so no base change is involved). Multiplication by 3 is a nonzero isogeny of the elliptic curve $\mathrm{Pic}^0 C$, hence surjective on $\overline{k}$-points, so there is a class $\delta$ with $3\delta = \mathcal{O}_C(1) - 3[P_0]$. On a genus-one curve with a rational point every degree-zero class has the form $[Q] - [P_0]$, so $\delta = [Q] - [P_0]$ for some $Q \in C(\overline{k})$, and then $3[Q] \sim \mathcal{O}_C(1)$: the tangent at $Q$ cuts $3Q$, so $Q$ is a flex. Given one flex $O$, part (2) identifies the flexes with $C[3]$. Changing the chosen flex translates this identification, so without a choice the flexes form a torsor under $\mathrm{Pic}^0(C)[3]$ and their number is $\#C[3](\overline{k})$.

(4) Combine (2) and (3) with the structure of $C[3]$ in characteristic $\neq 3$ to conclude that a smooth plane cubic has exactly 9 flexes. What does comparing this with (1) tell you about the intersection $C \cap V(H(F))$?

Solution. **Exactly 9 flexes. Each multiplicity in $C \cap V(H(F))$ is 1.** For $\mathrm{char}\, k \neq 3$ the multiplication-by-3 map is separable of degree 9, so $C[3](\overline{k}) \simeq (\mathbb{Z}/3)^2$ has exactly 9 elements. By (3) there are exactly 9 flexes. Comparing with (1): nine flexes appear in an intersection of total degree nine, so every intersection multiplicity is 1.

*Remark.* That is, $C$ meets its Hessian *transversally*, something the Bézout count alone could not tell us.

(5) Specialize to $C_d$. Compute $H(F)$ for $F = X^3 + Y^3 - dZ^3$ and deduce that, as subsets of $C_d(\overline{k})$,

$$C_d[3](\overline{k}) = (C_d \cap \{XYZ = 0\})(\overline{k}).$$

Write down all nine flexes.

Solution. $\boldsymbol{H(F) = -216\, d\, XYZ.}$ The Hessian matrix of $F = X^3 + Y^3 - dZ^3$ is $\mathrm{diag}(6X, 6Y, -6dZ)$, so

$$H(F) = -216\, d\, XYZ.$$

Since $d \neq 0$ and $216 = 2^3 3^3$ is invertible for $\mathrm{char}\, k \neq 2, 3$, we get $V(H(F)) = \{XYZ = 0\}$, and by (2), (4) the nine flexes are $C_d \cap \{XYZ = 0\}$. (In characteristic 2 the Hessian vanishes identically. There, (6) below and the analogue of the tangent computation of Problem 1.4(2) at

$Q_1, Q_2$ give the same nine flexes.) Explicitly, with $\delta \in \overline{k}$ a cube root of $d$:

$$\begin{aligned} Z = 0: &\quad (1:-1:0), \quad (1:-\zeta:0), \quad (1:-\zeta^2:0), \\ X = 0: &\quad (0:\delta:1), \quad (0:\zeta\delta:1), \quad (0:\zeta^2\delta:1), \\ Y = 0: &\quad (\delta:0:1), \quad (\zeta\delta:0:1), \quad (\zeta^2\delta:0:1). \end{aligned}$$

Three on each line, nine in all.

(6) Check (5) against the group law directly: use (2.4) to show that every point $P = (0 : Y_0 : Z_0) \in C_d(\overline{k})$ satisfies $[2]P = \ominus P$.

Solution. Put $X_0 = 0$ in (2.4). Then $Y_0 \neq 0$, since $(0:0:1) \notin C_d$, and the coordinates become

$$\big(-Y_0 \cdot Y_0^3 : 0 : Z_0(0 - Y_0^3)\big) = (-Y_0^4 : 0 : -Y_0^3 Z_0) = (Y_0 : 0 : Z_0),$$

which is $\ominus(0 : Y_0 : Z_0)$ by Problem 2.1(1). So $[2]P = \ominus P$, hence $[3]P = O$, confirming that the points of $C_d(\overline{k})$ on the line $X = 0$ are 3-torsion. And by the symmetry $X \leftrightarrow Y$, likewise those on $Y = 0$.

(7) Which of the nine flexes lie in $C_d(k)$? Treat the three lines separately, and say what happens for $k = \mathbb{Q}$.

Solution. **Over $\mathbb{Q}$: only $O$, unless $d = \delta^3$, when also $(0 : \delta : 1)$ and $(\delta : 0 : 1)$.** On $Z = 0$: $O$ is always rational, and $Q_1, Q_2$ are rational exactly when $\zeta \in k$, by Problem 1.4(1). On $X = 0$ and $Y = 0$: the flex $(0 : \delta : 1)$ is rational if and only if $\delta \in k$, i.e. if and only if $d \in (k^\times)^3$. The remaining four need $\zeta$ as well. So over $k = \mathbb{Q}$, where $\zeta \notin \mathbb{Q}$, there are only two possibilities: if $d$ is not a rational cube the only rational flex is $O$, and if $d = \delta^3$ then $C_d(\mathbb{Q})$ contains exactly three flexes,

$$O, \qquad (0:\delta:1), \qquad (\delta:0:1),$$

which by (6) form a subgroup isomorphic to $\mathbb{Z}/3\mathbb{Z}$ (the last two are inverse to each other). One can never see all nine over $\mathbb{Q}$.

(8) ($\star$) In characteristic three, $\#C[3](\overline{k})$ is 3 or 1 according as $C$ is ordinary or supersingular, so a smooth plane cubic has 3 or 1 flexes. How is that compatible with (1)?

Solution. **No contradiction: the nine are counted with multiplicity, here 3, 3, 3 or 9.** Part (1) rests on Proposition 2.4, which we stated only in characteristic 0 or $> 3$, so as it stands it is not available. The argument of (2) and (3), by contrast, never mentions the Hessian and is valid in every characteristic. It gives $\#C[3](\overline{k})$ flexes, which is 3 or 1 when $\operatorname{char} k = 3$ because [3] is then inseparable. In fact, for cubics the Hessian still finds the flexes in characteristic three. What breaks is the transversality of (4). Take $C$ in Weierstrass form $y^2 = x^3 + a_2x^2 + a_4x + a_6$. (By (3) $C$ has a flex, and a linear change of coordinates as in Problem 3.4 reaches this form. Then $H(F)$ changes by the square of the determinant, so where it vanishes does not depend on the coordinates.) In characteristic three,

$$H(F) = Z\,(a_2^2X^2 + a_2a_4XZ + a_4^2Z^2 - a_2Y^2),$$

and on $C$ the second factor equals $a_4^2 - a_2a_6 - a_2x^3$, which is minus the 3-division polynomial $\psi_3 = b_2x^3 + b_8 = a_2x^3 + a_2a_6 - a_4^2$ of $C$ in characteristic three. So the second factor vanishes exactly at the points of order 3 [Sil09, Exercise 3.7]. The intersection $C \cap V(H(F))$ thus consists of the flexes only, but with multiplicities: the line $Z = 0$ is the flex tangent at $O$ and meets $C$ there with multiplicity 3; if $a_2 \neq 0$ (the ordinary case) the conic meets $C$ in the two points of order 3, which are swapped by $y \mapsto -y$, so with multiplicity 3 each; if $a_2 = 0$ (the supersingular

case) $H(F) = a_4^2 Z^3$ with $a_4 \neq 0$, since $C$ is smooth, and all nine lie at $O$. The Hessian criterion for cubics really does fail in characteristic 2, where $H(F)$ vanishes identically (see (5)).

In Problem 2.8 you may use the following two results. Part (4) asks what the classification of automorphism groups in [Sil09, §III.10] predicts.

**Theorem 2.6** (Morphisms fixing $O$ [Sil09, Theorem III.4.8])**.** *A morphism of elliptic curves taking $O$ to $O$ is a group homomorphism.*

**Proposition 2.7** (Endomorphism rings [Sil09, §III.4])**.** *The endomorphism ring of an elliptic curve has no zero divisors.*

**Problem 2.8** (Complex multiplication, $\star$)**.**
Let $\alpha(X : Y : Z) = (X : Y : \zeta Z)$, a map defined over $k(\zeta)$.

(1) Show that $\alpha$ maps $C_d$ to itself and fixes $O$.

Solution. $F(X, Y, \zeta Z) = X^3 + Y^3 - d\zeta^3 Z^3 = F(X, Y, Z)$ since $\zeta^3 = 1$, so $\alpha$ carries $C_d$ into $C_d$. And $\alpha(1 : -1 : 0) = (1 : -1 : 0)$.

(2) Conclude that $\alpha$ is a group homomorphism, hence an automorphism of the elliptic curve $(C_d, O)$ over $k(\zeta)$.

Solution. $\alpha$ is a morphism of curves sending $O$ to $O$. By [Sil09, Theorem III.4.8] any such morphism is a group homomorphism. It is invertible (with inverse given by $\zeta^2$), so it is an automorphism of $(C_d, O)$.

(3) Show that $\alpha^2 + \alpha + 1 = 0$ in $\mathrm{End}(C_d \times_k k(\zeta))$, the ring of endomorphisms of $C_d$ defined over $k(\zeta)$.

Solution. $\alpha^3 = \mathrm{id}$ because $\zeta^3 = 1$, so $(\alpha - 1)(\alpha^2 + \alpha + 1) = \alpha^3 - 1 = 0$ in $\mathrm{End}(C_d \times_k k(\zeta))$. The endomorphism ring of an elliptic curve is a ring without zero divisors (degrees multiply), and $\alpha \neq 1$ since $\zeta \neq 1$. Hence $\alpha^2 + \alpha + 1 = 0$.

(4) Deduce that $\mathbb{Z}[\zeta] \subseteq \mathrm{End}(C_d \times_k k(\zeta))$, so every Selmer curve has complex multiplication. What does the classification of automorphism groups then predict for $j(C_d)$ and for $\#\,\mathrm{Aut}(C_d)$ and $\#\,\mathrm{Aut}(C_d \times_k k(\zeta))$?

Solution. $\boldsymbol{j(C_d) = 0}$**. For** $\boldsymbol{k = \mathbb{Q}}$**,** $\boldsymbol{\#\,\mathrm{Aut}(C_d) = 2}$ **and** $\boldsymbol{\#\,\mathrm{Aut}(C_d \times_{\mathbb{Q}} \mathbb{Q}(\zeta)) = 6}$**.** By (3) the subring of $\mathrm{End}(C_d \times_k k(\zeta))$ generated by $\alpha$ is a quotient of $\mathbb{Z}[T]/(T^2 + T + 1) = \mathbb{Z}[\zeta]$. It is torsion-free, because the endomorphism ring is [Sil09, §III.4], and every proper quotient of $\mathbb{Z}[\zeta]$ is finite. So $\mathbb{Z}[\zeta] \subseteq \mathrm{End}(C_d \times_k k(\zeta)) \subseteq \mathrm{End}(\overline{C}_d)$. In particular the endomorphism ring is strictly larger than $\mathbb{Z}$, which is complex multiplication. A curve admitting an automorphism of order three has $j = 0$. And by [Sil09, Theorem III.10.1], in characteristic $\neq 2, 3$ a curve with $j = 0$ has automorphism group of order 6 over a field containing $\zeta$, and order 2 otherwise. For $k = \mathbb{Q}$ we expect $\#\,\mathrm{Aut}(C_d) = 2$ and $\#\,\mathrm{Aut}(C_d \times_{\mathbb{Q}} \mathbb{Q}(\zeta)) = 6$.

*Remark.* Example 4.2 confirms this in `Magma`.

## 3. From Riemann–Roch to a Weierstrass equation

This section reproduces, for our family, the proof of [Sil09, Proposition III.3.1]: every elliptic curve is isomorphic to a curve in Weierstrass form.

Problem 3.4 is the longest problem here. Parts (1)–(3) go quickly once you see that Proposition 3.2 converts "pole order at $O$" into "multiplicity of $O$ in $h \cap C_d$". Part (5) is a two-line computation with

a surprising answer, and (8) is worth coming back to. Weierstrass equations and their coefficients $a_1, \dots, a_6$ are as in [Sil09, §III.1].

**Definition 3.1** (Riemann–Roch spaces)**.** For $D \in \operatorname{Div} C_d$,

$$\mathcal{L}(D) = \left\{ f \in k(C_d)^\times : \operatorname{div}(f) \geqslant -D \right\} \cup \{0\}, \qquad \operatorname{div}(f) \in \operatorname{Div} C_d.$$

This is a space of functions defined *over* $k$.

You may use the following two results without proof.

**Proposition 3.2** (Bézout)**.** *For a nonzero linear form $h$, the scheme-theoretic intersection $h \cap C_d$ is an effective divisor of degree 3 in $\operatorname{Div} C_d$. Over $\overline{k}$ it becomes a sum of three points of $C_d(\overline{k})$ counted with multiplicity. Consequently, since $X + Y$ cuts out $3O$ by Problem 1.4(2),*

$$\operatorname{div}\left(\frac{h}{X+Y}\right) = (h \cap C_d) - 3O.$$

**Proposition 3.3** (Riemann–Roch for genus one [Sil09, §II.5 and proof of Proposition III.3.1])**.** *For every $n \geqslant 1$ we have $\dim_k \mathcal{L}(nO) = n$. Since $C_d$ is geometrically integral, $\mathcal{L}(D) \otimes_k \overline{k} = \mathcal{L}(D_{\overline{k}})$, so the dimension is the same whether computed over $k$ or over $\overline{k}$.*

**Problem 3.4** (The Weierstrass model of $C_d$)**.**
Consider the rational functions in $k(C_d)$

$$w \coloneqq \frac{Z}{X+Y}, \qquad v \coloneqq \frac{X-Y}{X+Y}.$$

For parts (2)–(6) and (8) assume $\operatorname{char} k \neq 2$. Part (7) replaces $v$ by $X/(X+Y) = (1+v)/2$, which also makes sense in characteristic 2, and reaches a model valid whenever $\operatorname{char} k \neq 3$.

(1) Using Problem 1.4(1) and Proposition 3.2, compute $\operatorname{div}(w)$. Deduce that $w$ has a pole of order exactly 2 at $O$ and no other pole, so $w \in \mathcal{L}(2O)$. Where are its zeros, and what are they in the language of Problem 2.5?

Solution. **$\operatorname{div}(w) = Q_1 + Q_2 - 2O$. Its zeros are the flexes $Q_1$, $Q_2$.** By Problem 1.4(1), $Z \cap C_d = O + Q_1 + Q_2$ with $Q_1, Q_2 \neq O$. So

$$\operatorname{div}(w) = O + Q_1 + Q_2 - 3O = Q_1 + Q_2 - 2O.$$

Hence the only pole is at $O$, of order exactly 2. This last computation is an identity in $\operatorname{Div} \overline{C}_d$. In $\operatorname{Div} C_d$, when $\zeta \notin k$, the pair $\{Q_1, Q_2\}$ is a *single* closed point $\mathfrak{q}$ of degree 2 (see Warning 4.1), and the same statement reads $\operatorname{div}(w) = \mathfrak{q} - 2O$. Either way the degree is 0 and the pole order at $O$ is 2. Its zeros $Q_1, Q_2$ are, by Problem 2.5(5) and Problem 1.4(1), the nontrivial 3-torsion points of $C_d(\overline{k})$ on the line $Z = 0$.

(2) Show that the line $X - Y = 0$ does not pass through $O$. Deduce that $v$ has a pole of order exactly 3 at $O$, so $v \in \mathcal{L}(3O)$. Where are its zeros?

Solution. **At the three points of order two, on the line $X = Y$.** The form $X - Y$ takes the value $1 - (-1) = 2$ at $O$, which is nonzero since $\operatorname{char} k \neq 2$. So $O$ does not appear in $(X - Y) \cap C_d$ and $\operatorname{div}(v) = ((X - Y) \cap C_d) - 3O$ has a pole of order exactly 3 at $O$. Its zeros are the three points of $C_d(\overline{k})$ on the line $X = Y$, i.e. the nontrivial 2-torsion of Problem 2.1(2). In $\operatorname{Div} C_d$ these assemble into closed points of total degree 3.

*Remark.* In characteristic 2, $X - Y = X + Y$ vanishes at $O$. That is why (7) avoids $v$.

(3) Using Proposition 3.3, explain why $\{1, w\}$ is a basis of $\mathcal{L}(2O)$ and $\{1, w, v\}$ is a basis of $\mathcal{L}(3O)$.

SOLUTION. $\mathcal{L}(O)$ has dimension 1 by Proposition 3.3 and contains the constants, so $\mathcal{L}(O) = k$. Now $\dim \mathcal{L}(2O) = 2$ and $1, w \in \mathcal{L}(2O)$ are independent because $w \notin k$ (it has a pole). So $\{1, w\}$ is a basis. Similarly $\dim \mathcal{L}(3O) = 3$ and $v \in \mathcal{L}(3O) \setminus \mathcal{L}(2O)$ since its pole order is 3, so $\{1, w, v\}$ is a basis.

(4) The seven functions $1, w, v, w^2, wv, w^3, v^2$ all lie in $\mathcal{L}(6O)$. Why must they satisfy a non-trivial linear relation over $k$? List their pole orders at $O$. Which two collide, and why does that force the relation to involve both?

SOLUTION. **Pole orders $0, 2, 3, 4, 5, 6, 6$. Here $w^3$ and $v^2$ collide.** The seven functions lie in $\mathcal{L}(6O)$, which has dimension 6 by Proposition 3.3. Seven vectors in a 6-dimensional space are dependent. Their pole orders at $O$ are

$$1,\ w,\ v,\ w^2,\ wv,\ w^3,\ v^2 \quad\longmapsto\quad 0,\ 2,\ 3,\ 4,\ 5,\ 6,\ 6.$$

The first five have *distinct* pole orders, hence are linearly independent (a nontrivial combination has a pole of order equal to the largest one occurring). So any relation must involve $w^3$ or $v^2$. And since these are the only two with pole order exactly 6, a relation involving one must involve the other, with the leading terms cancelling. It also follows that the space of relations is exactly one-dimensional.

(5) Find the relation.

SOLUTION. $\boldsymbol{3v^2 + 1 = 4d\,w^3}$**.** On the chart $X + Y = 1$ we have $w = Z$ and $v = X - Y$, so $X = (1 + v)/2$, $Y = (1 - v)/2$, and

$$X^3 + Y^3 = \frac{(1+v)^3 + (1-v)^3}{8} = \frac{2 + 6v^2}{8} = \frac{3v^2 + 1}{4}.$$

The defining equation $X^3 + Y^3 = dZ^3$ therefore reads

$$3v^2 + 1 = 4d\,w^3. \tag{3.1}$$

(6) Let $E_1 \colon y_1^2 = x_1^3 - 432d^2$. Show that the relation of (5) says that there is a morphism $\varphi_1 \colon C_d \to E_1$ with $\varphi_1^* x_1 = 12dw$ and $\varphi_1^* y_1 = 36dv$. Write down $\varphi_1$ and its inverse explicitly.

SOLUTION. $\boldsymbol{\varphi_1^{-1}(x_1, y_1) = (36d + y_1 : 36d - y_1 : 6x_1)}$**.** Multiplying (3.1) by $432d^2$ gives

$$(36dv)^2 + 432d^2 = (12dw)^3,$$

so the functions $12dw$ and $36dv$ satisfy the equation of $E_1$, and this defines $\varphi_1$. Explicitly,

$$\varphi_1(X : Y : Z) = \left(\frac{12dZ}{X+Y},\ \frac{36d(X-Y)}{X+Y}\right), \qquad \varphi_1^{-1}(x_1, y_1) = (36d + y_1 : 36d - y_1 : 6x_1).$$

That $\varphi_1^{-1}$ lands on $C_d$ is a one-line check:

$$(36d + y_1)^3 + (36d - y_1)^3 = 2(36d)^3 + 216d\,y_1^2, \qquad d(6x_1)^3 = 216d(y_1^2 + 432d^2),$$

and $2 \cdot 36^3 = 216 \cdot 432 = 93312$.

(7) Where exactly did you use $\operatorname{char} k \neq 2$? Show that there is instead a morphism $\varphi_2$ with $\varphi_2^* x_2 = 3dw$ and $\varphi_2^* y_2 = 9dX/(X + Y)$ to

$$E_2 \colon y_2^2 - 9d\,y_2 = x_2^3 - 27d^2, \tag{3.2}$$

which is a Weierstrass equation valid whenever $\operatorname{char} k \neq 3$. Which of $a_1, \ldots, a_6$ are nonzero?

Solution. **Only $a_3 = -9d$ and $a_6 = -27d^2$.** The hypothesis $\operatorname{char} k \neq 2$ was used three times: in (2), where $X - Y$ takes the value 2 at $O$ (in characteristic 2, $X - Y = X + Y$ and $v = 1$ has no pole at all); to divide by 2 when solving for $X$ and $Y$ in terms of $v$; and, most seriously, to complete the square, which is what reaches the form $y^2 = x^3 + a_6$. For $\varphi_2$, work on the chart $X + Y = 1$, where $\varphi_2^* y_2 = 9dX$ and $X - 1 = -Y$, so

$$(\varphi_2^* y_2)^2 - 9d\,\varphi_2^* y_2 = 81d^2 X(X-1) = -81d^2 XY.$$

Meanwhile $X^3 + Y^3 = (X+Y)^3 - 3XY(X+Y) = 1 - 3XY$, and this equals $dw^3$, so $XY = (1 - dw^3)/3$ and

$$(\varphi_2^* y_2)^2 - 9d\,\varphi_2^* y_2 = -27d^2(1 - dw^3) = 27d^3 w^3 - 27d^2 = (\varphi_2^* x_2)^3 - 27d^2.$$

No division by 2 occurred, and the identity holds whenever $\operatorname{char} k \neq 3$. In the notation of [Sil09, §III.1] this is $a_1 = a_2 = a_4 = 0$, $a_3 = -9d$, $a_6 = -27d^2$.

*Remark.* This is exactly why the general Weierstrass equation carries the coefficients $a_1$ and $a_3$ that one is tempted to discard.

(8) ($\star$) The relation in (5) has no $w^2$, $wv$, $w$ or $v$ term, although a general Weierstrass equation allows all four. Explain why, using the automorphisms $\sigma = \ominus$ of Problem 2.1 and $\alpha$ of Problem 2.8.

Solution. From the definitions, $\sigma(X : Y : Z) = (Y : X : Z)$ and $\alpha(X : Y : Z) = (X : Y : \zeta Z)$ give

$$\sigma^* w = w, \quad \sigma^* v = -v, \qquad \alpha^* w = \zeta w, \quad \alpha^* v = v.$$

By (4) the space of relations among the seven monomials is one-dimensional. Both $\sigma^*$ and $\alpha^*$ multiply each monomial by a scalar, so they preserve that space and act on it by a scalar: the relation is a simultaneous eigenvector. It involves $w^3$ and $v^2$, and

$$\sigma^* w^3 = w^3, \quad \sigma^* v^2 = v^2, \qquad \alpha^* w^3 = w^3, \quad \alpha^* v^2 = v^2.$$

Both are fixed. So both eigenvalues are 1, and every monomial that is *not* fixed by both must have coefficient zero. Those are $v$ and $wv$ (negated by $\sigma^*$) and $w$, $w^2$, $wv$ (scaled by $\alpha^*$): precisely the four missing terms.

*Remark.* The shape of the Weierstrass equation is dictated by the automorphism group.

## 4. Worked examples in Magma

Most of you have never opened `Magma`, so this section is written as worked examples rather than exercises: read them, then re-run them yourself and change things. Every line of output below is reproduced from `Magma` V2.28-8 [BCP97], with whitespace tidied and one long type name elided. Three short variations to try on your own are collected at the end.

**Warning 4.1** (Closed points in `Magma`)**.** A closed point $P$ of $C_d$ is a point of the scheme $C_d$. Its residue field $k(P)$ is a finite extension of $k$, and $\deg P := [k(P) : k]$. The closed points of degree one are the rational points, and a closed point $P$ with $k(P)/k$ separable corresponds to an orbit of $\deg P$ points of $C_d(\overline{k})$ under $\mathrm{Gal}(k^{\mathrm{sep}}/k)$. The divisor group $\operatorname{Div} C_d$ is the free abelian group on the closed points [Poo06, §1.9], whereas [Sil09, §II.3] works with points of $C_d(\overline{k})$ and Galois-invariant divisors on $\overline{C}_d$. `Magma` follows the first convention: `Places` returns closed points, `Divisor` builds elements of $\operatorname{Div} C_d$, and `RiemannRochSpace` returns $\mathcal{L}(D)$ as a $k$-vector space.

Each example continues the previous one. Begin with

```
QQ := Rationals();
```

```
P2<X,Y,Z> := ProjectiveSpace(QQ,2);
d := 7;
C := Curve(P2, X^3 + Y^3 - d*Z^3);
O := C![1,-1,0];
```

The notation `P2<X,Y,Z>` names the coordinates so that we can write equations in them, and `C!` coerces a list of three numbers into a rational point of `C`. Typing `O := (1:-1:0);` does not work.

**Example 4.2** (Setting up, and a trap)**.** First, Problem 1.2:

```
IsNonSingular(C);  // true
Genus(C);          // 1
O;                 // (-1 : 1 : 0)
```

The last line is worth a pause. We asked for $(1:-1:0)$ and got $(-1:1:0)$: a point of $\mathbb{P}^2(\mathbb{Q})$ is defined only up to scaling, and Magma normalizes the representative it stores, here by $-1$.

Now ask for the Weierstrass model. The second return value `mp` is the isomorphism from `C` to `E`.

```
E, mp := EllipticCurve(C, O);
E;
// Elliptic Curve defined by y^2 - 9/49*y = x^3 - 27/2401 over Rational Field
jInvariant(E);                          // 0
E eq EllipticCurve([0,0,0,0,-432*d^2]);   // false
```

That `false` looks alarming, since Problem 3.4(6) says the answer should be $E_1\colon y_1^2 = x_1^3 - 432d^2$. It is not a contradiction: `eq` compares Weierstrass *coefficients*, and by [Sil09, Proposition III.3.1(b)] a Weierstrass model is unique only up to the changes of variables $x = u^2x' + r$, $y = u^3y' + su^2x' + t$. The right question is whether the curves are isomorphic, and against which model:

```
Eours := EllipticCurve([0,0,-9*d,0,-27*d^2]);
Eours;                          // y^2 - 63*y = x^3 - 1323
bool, iso := IsIsomorphic(Eours, E);
bool;                           // true
IsomorphismData(iso);           // [ 0, 0, 0, 1/7 ]
```

So $r = s = t = 0$ and $u = 1/7 = 1/d$: Magma found exactly the model $E_2$ of Problem 3.4(7), with $x_2$ and $y_2$ divided by $d^2$ and $d^3$. The two maps agree as well,

```
DefiningEquations(mp);   // [ 3/7*Z, 9/49*X, X + Y ]
```

against our $(3dZ : 9dX : X + Y)$, the ratios being $d^2$ and $d^3$ exactly as $u = 1/d$ demands. The moral: for elliptic curves, `eq` is almost never the question you mean to ask.

We can even watch this happen for all $d$ at once, by working over the function field $\mathbb{Q}(d)$ instead of over $\mathbb{Q}$:

```
Fd<D> := FunctionField(Rationals());
P2d<Xd,Yd,Zd> := ProjectiveSpace(Fd,2);
Cd := Curve(P2d, Xd^3 + Yd^3 - D*Zd^3);
Ed, mpd := EllipticCurve(Cd, Cd![1,-1,0]);
Ed;
```

```
// Elliptic Curve defined by y^2 - 9/D^2*y = x^3 - 27/D^4
DefiningEquations(mpd);
// [ 3/D*Zd, 9/D^2*Xd, Xd + Yd ]
```

which is (3.2) rescaled by $u = d$, as an identity in $d$.

Finally, the predictions of Problem 2.8(4):

```
#AutomorphismGroup(E);                    // 2
K<zeta> := CyclotomicField(3);
#AutomorphismGroup(BaseChange(E,K));      // 6
```

**Example 4.3** (The addition formulas, checked)**.** A `Magma` `function` takes arguments and returns a value. Here both are sequences of three coordinates, so that (2.3) and (2.4) can be typed almost verbatim.

```
dbl := function(P)
  x := P[1]; y := P[2]; z := P[3];
  return [-y*(2*x^3+y^3), x*(x^3+2*y^3), z*(x^3-y^3)];
end function;

add := function(P1,P2)
  A := P1[1]^2*P2[1] + P1[2]^2*P2[2] - d*P1[3]^2*P2[3];
  B := P1[1]*P2[1]^2 + P1[2]*P2[2]^2 - d*P1[3]*P2[3]^2;
  return [B*P1[2]-A*P2[2], B*P1[1]-A*P2[1], B*P1[3]-A*P2[3]];
end function;

P := [2,-1,1];
Q := dbl(P);      C!Q;   //  (5/3 : 4/3 : 1)
R := dbl(Q);      C!R;   //  (-1256/183 : 1265/183 : 1)
S := add(P,Q);    C!S;   //  (-17/38 : 73/38 : 1)
```

These are the rational points of Problem 2.3. Coercing through `C!` normalizes them, which is why $(15 : 12 : 9)$ prints as $(5/3 : 4/3 : 1)$. Now transport everything to $E$ through `mp` and compare with `Magma`'s own group law:

```
EP := mp(C!P);
mp(C!Q) eq 2*EP;          // true
mp(C!S) eq 3*EP;          // true
mp(C!R) eq 4*EP;          // true
```

Three `true`s: formulas derived from nothing but (2.1) and a $3 \times 3$ determinant agree with a general-purpose implementation.

**Example 4.4** (Riemann–Roch, mechanically)**.** `Magma` can carry out Problem 3.4 without being told the answer. The generators `x` and `y` of `FunctionField(C)` are the affine coordinates $X/Z$ and $Y/Z$ of $C_d$ (not Weierstrass coordinates), so our two functions are built from those.

```
FF<x,y> := FunctionField(C);    // x = X/Z,  y = Y/Z
w := 1/(x+y);                   // Z/(X+Y)
v := (x-y)/(x+y);               // (X-Y)/(X+Y)
pl0 := Place(0);
Valuation(w, pl0);              // -2
```

```
Valuation(v, pl0);              // -3
```

Those are the pole orders of Problem 3.4(1), (2). A valuation of $-2$ is a pole of order 2. As in Warning 4.1, `Place(0)` is the closed point $O$ and `Divisor` builds an element of $\operatorname{Div} C_d$, so everything here happens over $\mathbb{Q}$. Next, Proposition 3.3 itself:

```
D := Divisor(pl0);
for n in [1..6] do print n, Dimension(RiemannRochSpace(n*D)); end for;
// 1 1
// 2 2
// 3 3
// 4 4
// 5 5
// 6 6
```

Now ask for a basis of $\mathcal{L}(2O)$ without suggesting one:

```
V2, m2 := RiemannRochSpace(2*D);
[m2(b) : b in Basis(V2)];       // [ x^2 - y*x + y^2, 1 ]
```

which looks nothing like our $w$. But $X^3+Y^3=(X+Y)(X^2-XY+Y^2)$, and on $C_d$ the left side is $dZ^3$, so

$$\frac{X^2-XY+Y^2}{Z^2}=\frac{dZ^3}{Z^2(X+Y)}=d\,\frac{Z}{X+Y}=d\,w,$$

and indeed

```
(x^2 - x*y + y^2) eq d*w;       // true
```

Finally the relation. We list the seven functions of Problem 3.4(4), express each in a basis of $\mathcal{L}(6O)$ (that is what `@@` does, applying the inverse of the map `mV`), and take the kernel of the resulting matrix.

```
V, mV := RiemannRochSpace(6*D);
fns := [FF!1, w, v, w^2, w*v, w^3, v^2];
[-Valuation(f, pl0) : f in fns];   // [ 0, 2, 3, 4, 5, 6, 6 ]
M := Matrix([Eltseq(f @@ mV) : f in fns]);
K := Kernel(M);
Dimension(K);                      // 1
Basis(K);                          // [ ( 1  0  0  0  0  -28  3) ]
```

The pole orders are $0,2,3,4,5,6,6$: seven functions, six available orders, the collision at 6 forcing the relation exactly as in Problem 3.4(4). And the kernel is one-dimensional, spanned by the vector reading

$$1-28w^3+3v^2=0,$$

which is $3v^2+1=4dw^3$ at $d=7$. We got it by hand from a factorization. `Magma` got it from linear algebra in a Riemann–Roch space.

**Now try these.** Each one is a small edit to Example 4.2 or Example 4.3, and in each case the worksheet already tells you what the answer must be.

(1) Set `d := 2`. Problem 2.1(3) predicts a rational point of order two. Find it, verify with the group law that it has order two, and compute its image under $\varphi_1$ by hand.

Solution. **The rational point $(1:1:1)$, with $\varphi_1(1:1:1) = (12, 0)$.** $d = 2 = 2 \cdot 1^3$, so by Problem 2.1(3) the rational 2-torsion point is $(1:1:1)$, and indeed $1^3 + 1^3 = 2$. Then

$$\varphi_1(1:1:1) = \left(\frac{12 \cdot 2 \cdot 1}{1+1},\ \frac{36 \cdot 2 \cdot 0}{1+1}\right) = (12,\, 0),$$

which lies on $E_1 \colon y_1^2 = x_1^3 - 432 \cdot 4 = x_1^3 - 1728$ because $12^3 = 1728$. A point of $E_1(\overline{k})$ with $y_1 = 0$ is exactly a point of order two. Equivalently `dbl([1,1,1])` returns $(-3:3:0) = O$.

(2) Set `d := 8`. Problem 2.5(7) predicts rational 3-torsion. Find the two rational 3-torsion points other than $O$, use your `dbl` function to check that $[2]P = \ominus P$, and compute $\varphi_1(P)$ by hand.

Solution. **$(2:0:1)$ and $(0:2:1)$, with $\varphi_1(2:0:1) = (48, 288)$.** $d = 8 = 2^3$, so $\delta = 2$ and the rational flexes are $(2:0:1)$ and $(0:2:1)$ besides $O$. Doubling: `dbl([2,0,1])` gives

$$\big(-0 : 2(8+0) : 1(8-0)\big) = (0:16:8) = (0:2:1),$$

which is $\ominus(2:0:1)$, so $[3](2:0:1) = O$. By hand,

$$\varphi_1(2:0:1) = \left(\frac{12 \cdot 8 \cdot 1}{2},\ \frac{36 \cdot 8 \cdot 2}{2}\right) = (48,\, 288),$$

and $48^3 - 432 \cdot 64 = 110592 - 27648 = 82944 = 288^2$.

(3) Use `IsIsomorphic` to decide whether $C_1 \simeq C_8$ and whether $C_1 \simeq C_2$ over $\mathbb{Q}$. Pass it the *Weierstrass* models: on the plane curves themselves Magma answers `For genus 0 or 1 curves the basefield must be finite`. Reconcile the answers with the cubic-twist criterion of Section 5.

Solution. **$C_1 \simeq C_8$ over $\mathbb{Q}$, but $C_1 \not\simeq C_2$.** Over $\mathbb{Q}$ the curves $y_1^2 = x_1^3 + A_1$ and $y_2^2 = x_2^3 + A_2$ are isomorphic exactly when $A_1/A_2$ is a sixth power. For $C_1$ and $C_8$ we get $-432/(-432 \cdot 64) = 1/64 = (1/2)^6$, so `IsIsomorphic(EllipticCurve(C1,O1), EllipticCurve(C8,O8))` returns `true`. Consistently, $1/8 = (1/2)^3$ is a cube. For $C_1$ and $C_2$ we get $1/4$, which is not a sixth power in $\mathbb{Q}^\times$ (that would need $2^{-1/3} \in \mathbb{Q}$), so the answer is `false`. Consistently, 2 is not a rational cube. This is the criterion of Section 5: $C_d \simeq C_{d'}$ over $k$ if and only if $d/d' \in (k^\times)^3$.

## 5. Where the name comes from

Selmer [Sel51] studied the general diagonal cubic $aX^3 + bY^3 + cZ^3 = 0$, and produced the standard counterexample to the Hasse principle: the curve $3X^3 + 4Y^3 + 5Z^3 = 0$ has real points and $\mathbb{Q}_p$-points for every prime $p$, but no rational point. Such a curve is a torsor under its Jacobian [Poo17, §5.12], and that Jacobian is a Selmer curve in our sense:

$$\mathrm{Jac}(aX^3 + bY^3 + cZ^3) \simeq \big(y^2 = x^3 - 432(abc)^2\big).$$

Two ways to test this in Magma: the curve $X^3 + 2Y^3 + 3Z^3 = 0$ *does* have the rational point $(1:1:-1)$, so it is its own Jacobian and should be isomorphic to $C_6$, that is to $y^2 = x^3 - 432 \cdot 6^2$; and for Selmer's own curve one can call `Jacobian(GenusOneModel(3*X^3 + 4*Y^3 + 5*Z^3))` and compare with $y^2 = x^3 - 432 \cdot 60^2$. Both come out `true`.

Finally, by Problem 3.4(6) together with the classification of twists of $y^2 = x^3 + A$ [Sil09, §X.5], for $\mathrm{char}\, k \neq 2, 3$ the curves $C_d$ and $C_{d'}$ are isomorphic over $k$ exactly when $d/d' \in (k^\times)^3$, so the family is parametrized by $k^\times/(k^\times)^3$. These are the cubic twists of $X^3 + Y^3 = Z^3$, and they are the natural first examples for the descent machinery of [Sil09, §X] and [Poo17, Ch. 8].

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